

1. Administrative & Study Identification

Full Study Title: REAL-CARE: Real-world Effectiveness of Iptacopan in Italian Patients With Paroxysmal Nocturnal Hemoglobinuria (REAL-CARE) **Short Title/Acronym:** REAL-CARE **Protocol Number:** [Insert Official Protocol Title/Novartis Internal ID] **NCT Number:** NCT07229235 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Iptacopan (Factor B inhibitor) **Phase of Development:** Observational / Phase IV (Real-World Evidence) **Medical Monitor:** Novartis Clinical Operations Lead / [Name and Contact Information]

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the long-term hematological response following iptacopan initiation in routine clinical practice. **Secondary Objectives:** To evaluate the proportion of patients remaining free from red blood cell (RBC) transfusions, overall safety and tolerability, and changes in patient-reported outcomes including health-related quality of life (PROMIS-29+2), work productivity (WPAI:SHP), and medication adherence (MARS-5).

2.2 Background & Rationale Paroxysmal Nocturnal Hemoglobinuria (PNH) is a rare, life-threatening blood disorder characterized by the destruction of red blood cells (hemolysis) via an overactive complement system, leading to severe anemia, fatigue, and reliance on blood transfusions. Iptacopan is a novel oral factor B inhibitor that targets the alternative complement pathway. Unlike standard anti-C5 therapies, iptacopan offers a new mechanism of action by blocking an earlier step in the cascade, providing the potential for more complete protection against red blood cell destruction. This REAL-CARE study aims to generate real-world evidence confirming iptacopan's effectiveness and safety in a routine clinical practice setting for Italian patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Real-world effectiveness study) **Control Type:** Single-arm (Routine medical care) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: [Total number of evaluable patients required] **Number of Sites:** Italian clinical sites **Estimated Study Start:** [DD-MMM-YYYY] **Estimated Primary**

Completion: Patients will be followed up for 24 months post-initiation.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female, ≥ 18 years of age. 2. Documented diagnosis of Paroxysmal Nocturnal Hemoglobinuria (PNH). 3. Followed in Italian clinical sites, on treatment with iptacopan prescribed as per routine medical care (i.e., patients naïve to treatment for whom the decision to start has already been made based on clinical practice and SmPC/AIFA criteria, or patients already treated under the Managed Access Program). 4. Signed informed consent (unless deceased, as per Italian local regulations for retrospective data).

4.2 Exclusion Criteria 1. Current participation in an interventional clinical trial that prohibits observational data collection. 2. Inability to comply with standard of care monitoring and data collection required for real-world assessment. 3. [Additional protocol-specific exclusions, e.g., missing baseline laboratory values].

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Iptacopan dose prescribed as per standard routine medical care and SmPC recommendations. **Route of Administration:** Oral **Duration of Treatment:** Patients will be followed up for 24 months or until iptacopan discontinuation (due to adverse events, lack of efficacy, clinical decision, death, etc.).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care treatments for PNH and associated comorbidities. Patients may be treated with iptacopan as a monotherapy or switched from previous anti-C5 therapies. **Prohibited:** Use of unapproved experimental agents outside of the specified PNH routine care framework.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Index Date	Day 0	Iptacopan Initiation, Baseline Hemoglobin, Informed Consent
Follow-up 1	Day 14	
Month 12	Month 12	Track RBC Transfusions, AEs, Hematological Response
Interim	Month 12	Primary Endpoint Assessment (Hb levels), PRO Questionnaires (PROMIS-29+2, WPAI:SHP, MARS-5)
End of Study	Month 24	Final Observation, Long-term Effectiveness and Safety Assessments, Exit Data Collection

(Note: Data is collected as part of routine clinical visits rather than strictly scheduled interventional trial visits).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Absolute change in hemoglobin (Hb) levels at 12 months post-initiation of iptacopan.

Measurement Method: Clinical and laboratory records extraction.

7.2 Secondary & Exploratory Endpoints 1. Proportion of patients who remain free from red blood cell (RBC) transfusions from Day 14 through Month 12. 2. Absolute change from iptacopan initiation in generic health-related quality of life, evaluated via the PROMIS-29+2 score (covering 9 health domains including fatigue, pain, and sleep). 3. Change in work productivity and daily activities assessed by the WPAI:SHP questionnaire (absenteeism, presenteeism, and overall work productivity loss). 4. Patient-reported adherence to treatment assessed via the MARS-5 questionnaire.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine clinical monitoring for safety signals, including any AEs leading to the discontinuation of iptacopan.

Serious Adverse Events (SAEs): Evaluated and reported per real-world pharmacovigilance protocols and regulatory requirements.

Data Monitoring Committee (DMC): Overseen by the Sponsor's safety review committees for real-world evidence data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Patients successfully indexed on the date of iptacopan initiation who meet all eligibility criteria.

Sample Size Justification: Powered descriptively to capture real-world statistical significance in Hb level shifts within the Italian PNH population.

Handling of Missing Data: Standard real-world evidence (RWE) methodologies will be applied, including multiple imputation where sufficient baseline characteristics are available, recognizing the un-scheduled nature of routine care testing.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: The study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP), local Italian data protection laws (e.g., Legislative Decree n° 196), and the Declaration of Helsinki.

Monitoring Plan: Retrospective and prospective Electronic Health Record (EHR) and patient chart data extraction. **Audit Trail:** Secure database entry utilizing

anonymized patient identification, complying with standard data privacy provisions for scientific research.

11. Study Identifiers & Governance

Primary ID: REAL-CARE **Registry Number:** NCT07229235 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** REAL-CARE: Real-world Effectiveness of Iptacopan in Italian Patients With Paroxysmal Nocturnal Hemoglobinuria **Official Regulatory Title:** REAL-CARE: Real-world Effectiveness of Iptacopan in Italian Patients With Paroxysmal Nocturnal Hemoglobinuria

12. Clinical Framework (PNH Specific)

Primary Condition: Paroxysmal Nocturnal Hemoglobinuria (PNH) **Diagnosis Threshold:** Documented clinical diagnosis of PNH **Medical Methodology:** Standard of care utilization of factor B inhibitor **Optimization Target:** Hemoglobin (Hb) normalization and RBC transfusion independence

13. Study Procedures & Methodology

Management Pathway: Routine medical care prescribing of Iptacopan **Education Protocol (HCP):** N/A for observational setting; based on approved SmPC **Education Protocol (Patient):** N/A; standard clinic instruction and patient-reported outcome questionnaire training **Quality Audit Mechanism:** Assessment of self-reported adherence via the MARS-5 questionnaire

14. Observation & Follow-Up Schedule

Total Duration: Up to 24 months **Onsite Visit Cadence:** Dictated by routine clinical practice **Remote Monitoring Frequency:** N/A (EHR extraction) **Checklist Review Interval:** Routine intervals corresponding with WPAI:SHP and PROMIS-29+2 administration

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				